

Account

Account

1. Feature Overview

Open accounts under a customer, enforce account policies (opening deposit / min balance metadata), search accounts, update status, and post simple deposits that credit balances.

Status: Implemented for open / search / status / deposit / policies. **No** transaction ledger, withdraw, or transfer.

2. Business Purpose

Hold customer balances and product types (SAVINGS, CURRENT, SALARY, FD, RD, LOAN) with branch-coded account numbers.

3. User Workflow

- 1. /app/accounts — search list, Deposit action
- 2. Customer detail — Add Current (loads policy → opening deposit dialog if required) / Add Loan
- 3. Customer create — optional first account
- 4. Policies seeded at startup; create policy via API

4. Execution Flow

See [CALL FLOW.md](#), [deposit](#), [get active account policy](#).

5. Database Tables

Table	Purpose
account	Balances, status, number parts
account_policy	Per type+currency rules
account_ordinal_seq	Ordinal for number generation
customers	Owner FK

6. REST APIs

Method	Path	Roles
POST	/accounts	ADMIN, EMPLOYEE
GET	/accounts	ADMIN, EMPLOYEE, MANAGER
GET	/accounts/customer/{customerId}	ADMIN, EMPLOYEE, MANAGER
PUT	/accounts/{accountId}/status	ADMIN, EMPLOYEE
POST	/accounts/{accountId}/deposit	ADMIN, EMPLOYEE

Method	Path	Roles
POST	/account-policies	Authenticated*
GET	/account-policies?accountType¤cyCode	Authenticated*

*Restrict to ADMIN — deferred hardening.

7. Controllers

- AccountController
- AccountPolicyController

8. Services

- AccountService / AccountServiceImpl — openAccount, getAccountsByCustomerId, updateAccountStatus, deposit, searchAccounts
- AccountPolicyService / AccountPolicyServiceImpl — createPolicy, getActivePolicy
- AccountPolicyInitializer — seed INR policies

9. Repositories

- AccountRepository (+ getNextOrdinal() native query)
- AccountPolicyRepository

10. DTOs

OpenAccountRequest, AccountResponse, UpdateAccountStatusRequest, DepositRequest, CreateAccountPolicyRequest, AccountPolicyResponse

11. Entities / Enums

- Account, AccountPolicy
- AccountType, AccountStatus, CurrencyCode

12. Utility Classes

- AccountNumberGenerator.generate(...)
- AccountSpecification.matching(...)

13. Configuration Classes

SecurityConfig matchers for /accounts/**; policy paths fall under anyRequest().authenticated().

Frontend: core/services/account.ts, account-policy.ts, features/accounts/, features/opening-deposit-dialog/

14. Validation Rules

- Open: customer exists; active policy in effective window; if openingDepositRequired then amount $\geq$ requiredOpeningDeposit
- Deposit: account ACTIVE; amount positive
- Policy create: Bean Validation on CreateAccountPolicyRequest

15. Security Rules

Account mutations ADMIN/EMPLOYEE; reads include MANAGER. Policy endpoints currently any authenticated user.

16. Exception Handling

Business exceptions for missing account/customer/policy and validation failures $\rightarrow$ GlobalExceptionHandler.

17. Logging

SQL debug in dev; no structured account event log.

18. Audit Events

Account has created_by / timestamps; AccountPolicy uses AuditableEntity. No immutable audit trail of deposits.

19. Testing Strategy

- Open CURRENT with deposit $< 5000 \rightarrow$ reject
- Open CURRENT with $\geq$ policy min $\rightarrow$ balances equal deposit
- Deposit on ACTIVE $\rightarrow$ balances and creditCount increase
- Deposit on non-ACTIVE $\rightarrow$ reject
- GET policy CURRENT/INR returns seed row
- UI: Add Current opens dialog when required

20. Future Extension Guide

- Persist ledger rows on every credit/debit
- Withdraw / transfer with limits
- Account detail page
- Status transition state machine
- Seed LOAN policy or block LOAN until product module exists
- Maker-checker for large deposits

Future Modification Guide

Requirement: Increase required opening deposit for CURRENT/INR

Item	Detail
Prefer	Update/insert account_policy row (API POST /account-policies or DB) — not hardcode UI
Files	AccountPolicyInitializer.java (seed), or runtime policy via AccountPolicyServiceImpl.createPolicy
Classes	AccountPolicy, AccountPolicyServiceImpl
Methods	createPolicy(), getActivePolicy(); enforced in AccountServiceImpl.openAccount()
Impact	OpeningDepositDialog min; customer onboarding amounts

Requirement: Change deposit to write a ledger entry

Item	Detail
Files	New Transaction module + AccountServiceImpl.deposit / openAccount
Methods	deposit() must call transaction create after balance update (or same TX)
Impact	Dashboard todayTransactionCount; Reports; Audit

Requirement: Change account number format

Item	Detail
Files	AccountNumberGenerator.java, possibly AccountServiceImpl.openAccount
Methods	generate(branchCode, accountType, ordinal, currency, checkDigit)
Impact	Display/search only for new accounts; old numbers unchanged

Requirement: Allow deposit on FROZEN accounts

Item	Detail
Files	AccountServiceImpl.java
Methods	deposit() status check
Impact	Status rules doc; UI Deposit button enablement

Call hierarchy (open account)

```

AccountController.openAccount()
    ↓
AccountServiceImpl.openAccount()
    ↓
CustomerRepository.findById()
    ↓
AccountPolicyRepository.findByAccountTypeAndCurrencyCodeAndActiveTrue()
    ↓
AccountRepository.getNextOrdinal()
    ↓
AccountNumberGenerator.generate()
    ↓
AccountRepository.save()

```

Call hierarchy (deposit)

```
AccountController.deposit()  
  ↓  
AccountServiceImpl.deposit()  
  ↓  
AccountRepository.findById()  
  ↓  
AccountRepository.save()
```